

Land Surface Temperature Prediction in Egypt Using Machine Learning

Project Report

Project scope: Machine-learning regression using historical ERA5 hourly meteorological data for a location in Giza, Egypt (2000–2025).

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1. Project Overview

This project investigates whether traditional machine-learning regression models can predict land surface temperature from historical meteorological and temporal information. The target variable is land surface temperature, selected because temperature is an important environmental indicator with implications for agriculture, water resources, and human well-being.

The project uses hourly ERA5 reanalysis data from the Copernicus Climate Data Store for a location in Giza, Egypt. Three regression approaches are compared: Linear Regression, XGBoost, and Random Forest. The models are evaluated using a chronological train/test split and three-fold cross-validation.

The final experiments identify Random Forest as the strongest of the tested models. Hyperparameter tuning does not improve the held-out test score, but it substantially reduces the gap between training and testing performance, providing a less overfit model with similar generalization performance.

2. Problem Definition

The objective is to build a regression model that estimates land surface temperature from meteorological variables and calendar information. Rather than treating the project as a complete climate-change forecasting system, this work is best understood as a machine-learning prediction study on a historical time series.

The prediction target is land surface temperature (`t2m` in the source dataset). The model uses surface pressure, precipitation, wind characteristics, and temporal features such as hour, month, quarter, day of week, and year. The project also compares several model families to determine whether nonlinear ensemble methods provide an advantage over a simple linear baseline.

3. Dataset

The dataset consists of ERA5 hourly time-series data obtained from the Copernicus Climate Data Store. The geographic reference is approximately 29.84178° N, 30.65322° E in Giza, Egypt. The dataset covers January 2000 through May 2025. The original processed CSV contains 222,072 observations and nine variables. No missing values were detected in the supplied dataset.

Variable	Description	Use
valid_time	Observation timestamp	Datetime index
u10	10 m eastward wind component	Used to derive wind speed/direction
v10	10 m northward wind component	Used to derive wind speed/direction
t2m	2 m temperature	Target variable
msl	Mean sea-level pressure	Removed
sp	Surface pressure	Predictor
tp	Total precipitation	Predictor
latitude / longitude	Location coordinates	Removed

4. Data Preprocessing & Feature Engineering

The timestamp was converted to a pandas datetime index. The dataset was checked for missing values and no null values were present. A drop-and-reset operation was retained in the notebook as a defensive preprocessing step.

Temperature was converted from Kelvin to Celsius by subtracting 273.15. Total precipitation was converted from meters to millimeters by multiplying by 1,000. Mean sea-level pressure was removed because it was visually very similar to surface pressure in the exploratory analysis. Latitude and longitude were also removed because the dataset represents a single fixed geographic location.

Calendar features were extracted from the timestamp: hour, day of week, month, quarter, and year. The wind components were transformed into wind speed using the magnitude of the two components and into wind direction using the project's directional calculation. The final modeling dataset contains the following features: hour, day of week, month, quarter, year, wind speed, wind direction, surface pressure, and precipitation. Land surface temperature is the target.

5. Methodology

The data was divided chronologically rather than randomly. Observations before 1 January 2020 were used for training, while observations from 1 January 2020 onward were held out for testing. This preserves the temporal ordering of the historical data and provides a more realistic evaluation than a random split.

Three regression models were evaluated: Linear Regression as a simple baseline, XGBoost as a nonlinear gradient-boosting method, and Random Forest as a nonlinear tree ensemble. Model performance was evaluated using R^2 , Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Three-fold cross-validation was also applied to the training data to examine the stability of model performance.

6. Model Results

The initial experiments show a clear performance difference between the three models. Linear Regression achieved a test R^2 of 0.603, while XGBoost achieved 0.770. Random Forest performed best, achieving a test R^2 of 0.911, MAE of 1.702, and RMSE of 2.250. The results indicate that the relationship between the selected predictors and land surface temperature is substantially nonlinear, since the tree-based ensemble models outperform the linear baseline.

Model	CV mean R^2	Test R^2	Test MAE	Test RMSE
Linear Regression	0.568	0.603	3.769	4.739
XGBoost	0.757	0.770	2.852	3.610
Random Forest	0.916	0.911	1.702	2.250

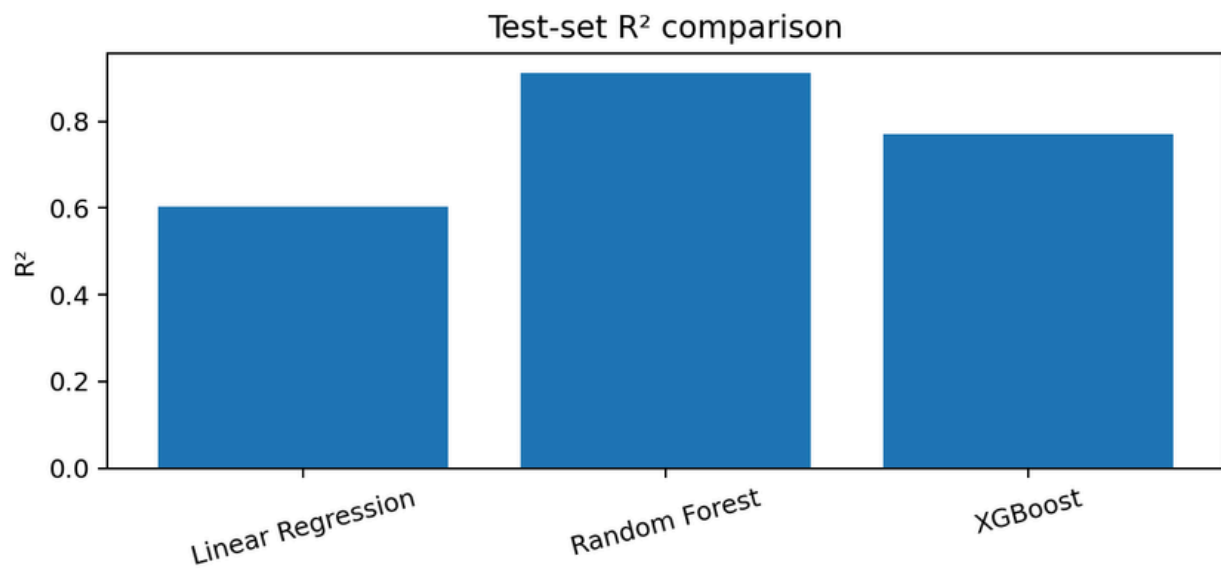


Figure 1. Test-set R^2 comparison.

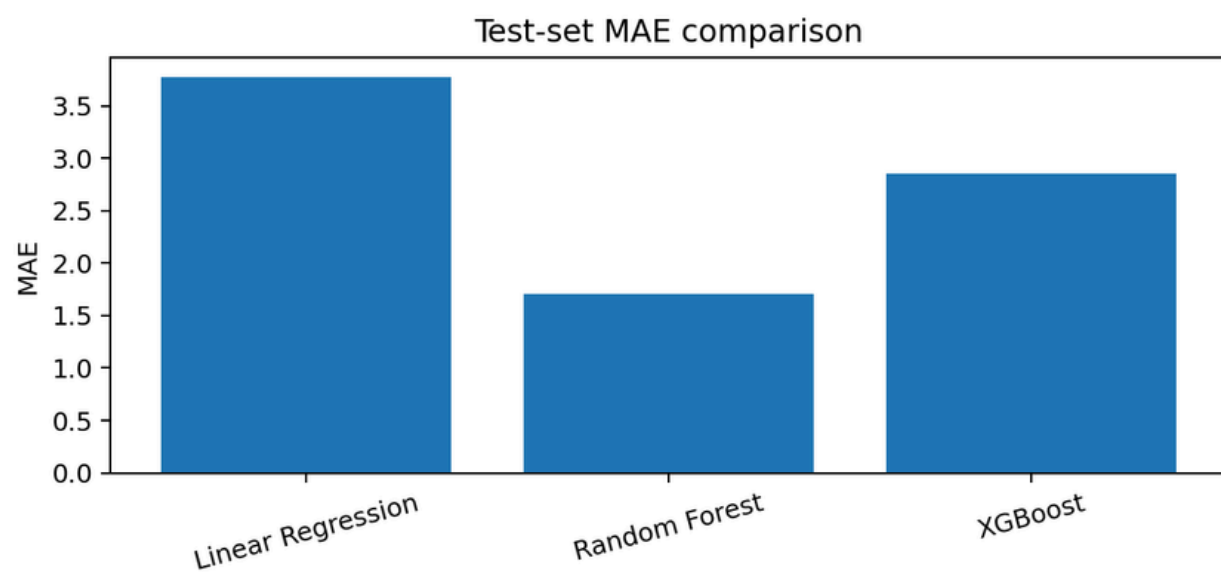


Figure 2. Test-set MAE comparison.

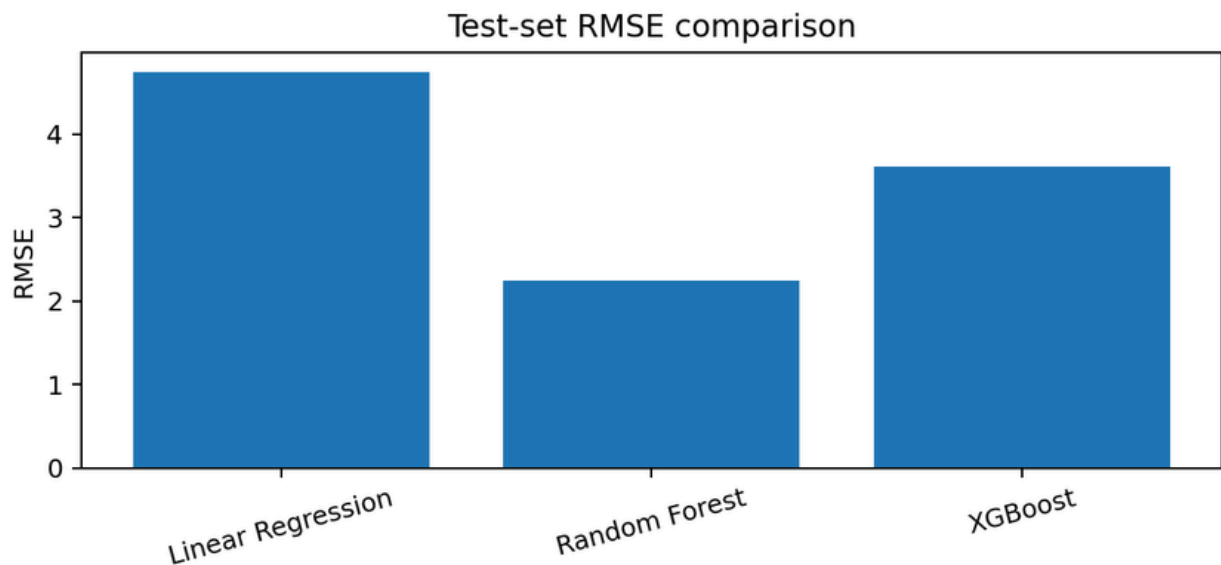


Figure 3. Test-set RMSE comparison.

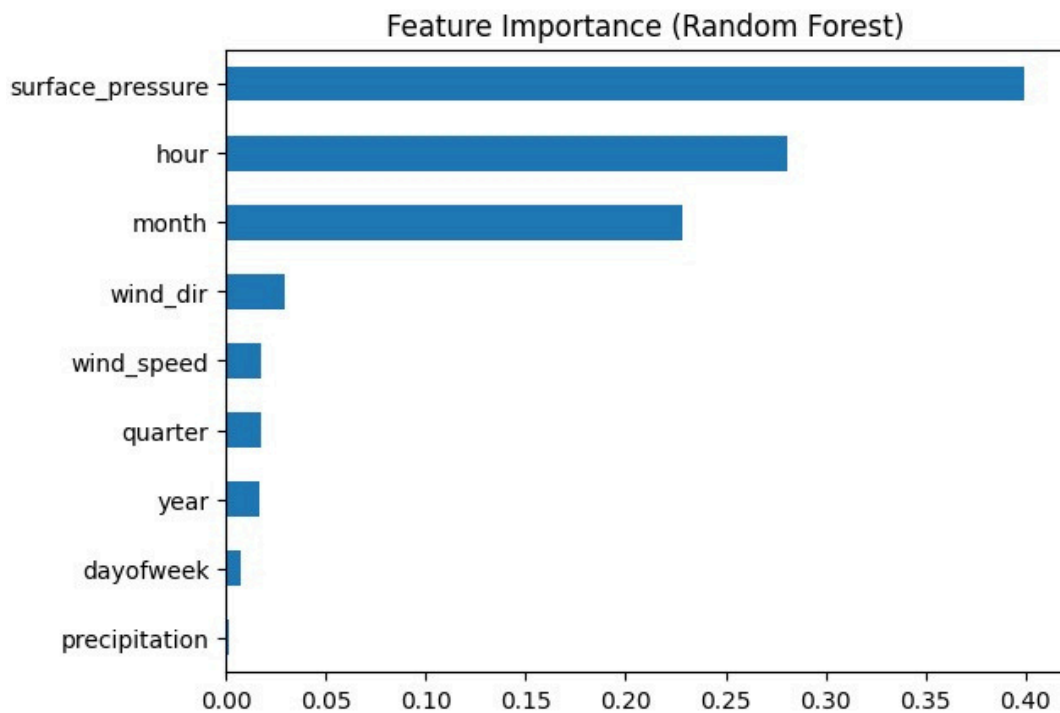
7. Hyperparameter Tuning

The initial Random Forest showed a large difference between training and testing performance: training R^2 was 0.995 compared with 0.911 on the test set. This indicates substantial overfitting. A grid search was performed over the number of estimators, maximum tree depth, minimum samples for splits and leaves, and the number of features considered at each split. The selected configuration used 100 estimators, a maximum depth of 10, ``sqrt`` max features, a minimum leaf size of 2, and a minimum split size of 2.

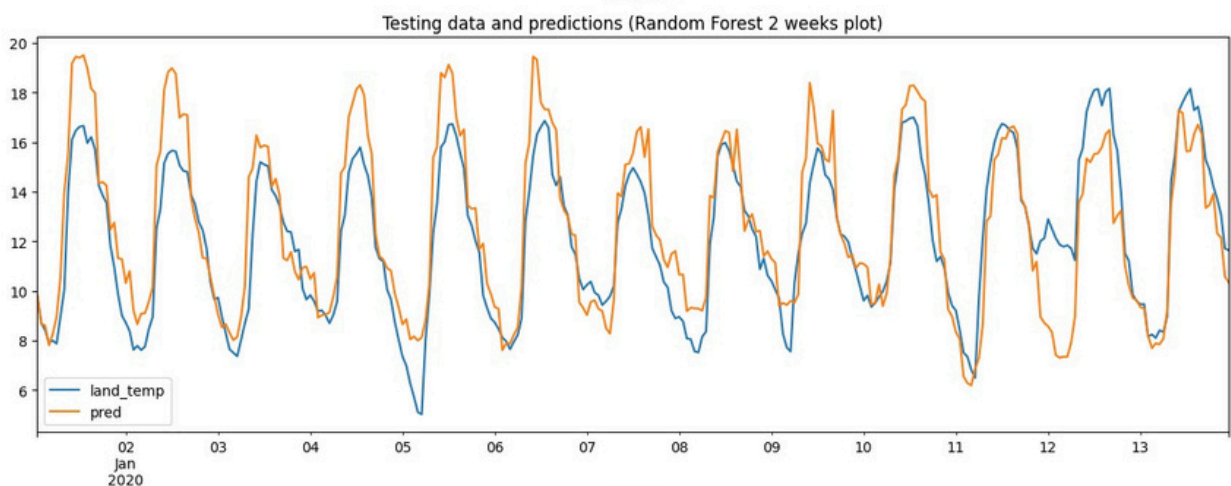
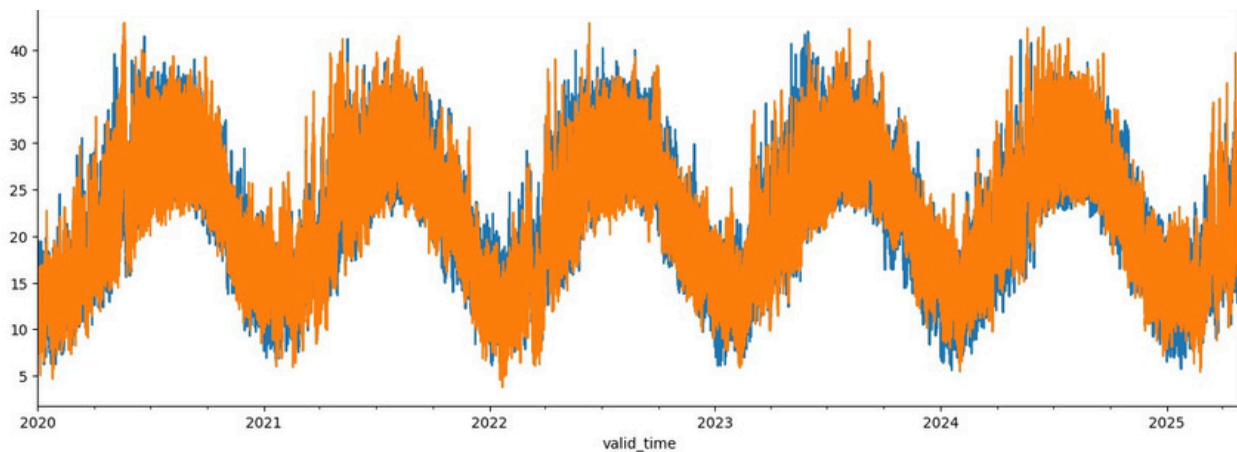
The tuned model achieved a cross-validation R^2 of 0.9045 ± 0.0059 , training R^2 of 0.922, and test R^2 of 0.907. Therefore, tuning did not improve the held-out test score, which decreased slightly from 0.911 to 0.907. However, it reduced the training/test gap considerably, indicating lower overfitting while retaining similar generalization performance.

8. Feature Importance & Prediction Analysis

The Random Forest feature-importance analysis indicates that surface pressure and hour of day are among the most influential predictors, followed by temporal features such as month. Precipitation has relatively low importance compared with the other selected features. These feature-importance values should be interpreted as model-specific importance rather than direct evidence of physical causation or statistical correlation. In particular, a feature being important to the Random Forest does not by itself establish that it causes temperature changes.



The prediction plots in the notebook show that the Random Forest generally follows the observed temperature pattern in the test period, although there are intervals where predictions do not fully capture the observed variation. This suggests useful generalization while also showing that prediction errors remain.



9. Discussion

Random Forest was the strongest model among the three approaches tested. Its nonlinear structure is well suited to relationships between temperature and meteorological variables that may not be adequately represented by a linear model. The tuned Random Forest is arguably the more defensible final model for presentation because it provides similar test performance while reducing the extreme training performance of the original model. The difference between the tuned model's cross-validation score and held-out test score is also relatively small, supporting reasonably stable generalization within this dataset. However, the model should be interpreted as a baseline predictive study rather than a complete operational forecasting system. Its performance is specific to the dataset, location, feature set, and evaluation design used in this project.

10. Limitations

- **Single location:** The dataset represents one geographic location in Giza. The model should not be assumed to generalize to other parts of Egypt without additional spatial data and validation.
- **No lagged temperature features:** The model does not explicitly use previous temperature observations such as 1-hour, 24-hour, or 7-day lags. These could be important predictors because temperature is temporally autocorrelated.
- **Forecasting horizon:** The current experiment evaluates predictions on a later historical period but does not define a specific operational forecast horizon such as 1 hour ahead or 24 hours ahead.
- **Feature representation:** Hour, month, and direction are represented as ordinary numeric variables. Cyclical encodings could be explored in a future version, especially for variables such as hour and wind direction.
- **Model interpretation:** Feature importance describes how the Random Forest uses the available predictors; it should not be interpreted as proof of physical causality.

11. Future Improvements

Future work could introduce lagged temperature and meteorological variables, rolling statistics, and explicitly defined forecast horizons. Time-series-specific validation such as expanding-window or rolling-window evaluation could provide a stronger assessment of forecasting performance.

Additional locations could be incorporated to test spatial generalization across Egypt. More advanced models, including gradient boosting with systematic tuning or dedicated time-series/deep-learning approaches, could then be compared against the Random Forest baseline.

A production-oriented extension could package preprocessing and prediction into a reusable pipeline and expose the model through an API or small interactive application.

12. Conclusion

This project demonstrates a complete baseline machine-learning workflow for land surface temperature prediction using historical ERA5 data: data inspection, preprocessing, feature engineering, chronological evaluation, model comparison, hyperparameter tuning, and interpretation. Among Linear Regression, XGBoost, and Random Forest, Random Forest produced the strongest initial performance, with a test R^2 of approximately 0.911, MAE of 1.702, and RMSE of 2.250. Hyperparameter tuning reduced overfitting while maintaining a similar test R^2 of approximately 0.907.

Overall, the results support Random Forest as the most suitable model among the approaches tested for this dataset. The project also establishes a useful baseline for future work involving lagged features, stronger time-series validation, multiple locations, and deployment.